

Project Title: Deep Learning for Satellite-Based Wildfire Detection

Project Category (in the form of Method + Application, e.g. CNN for Face Recognition):

Convolutional Neural Networks (CNNs) for Wildfire Detection

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Repository Link: <https://github.com/isaactabor21/ds6050-final-project/>

I. Introduction/Motivation:

The goal of our project is to help address the problem of wildfire detection by building a deep learning model capable of identifying wildfire-affected regions from satellite imagery, by classifying satellite image patches as either wildfire or no wildfire. This is primarily an application result.

According to NASA, extreme wildfire activity has more than doubled worldwide, and has been amplified due to Earth's warming climate, particularly in northern and temperate forests, and satellite data is rich in helping detect and track them, by helping communities and land managers prepare for and respond to fires and understand this growing risk. NASA's Terra and Aqua satellites detect active wildfires twice each day, primarily by identifying thermal anomalies. We believe computer vision, particularly with CNNs, also has the potential to effectively support identifying wildfires through learning visual patterns, reducing false alarms, and improving accuracy. This could lead to timelier and more scalable environmental management and disaster response than manually analyzing large volumes of satellite imagery and relying on thermal sensors. In 2023, Canada's warmest and driest conditions since 1980 stoked extreme fires that lasted for five months. NASA researchers found that these fires released about 640 million metric tons of carbon.

II. Hypothesis:

We can train an effective Convolutional Neural Network (CNN) model to classify image patches as "fire" or "no fire" to enhance early detection of wildfires. We can outperform the model(s) built in Spiller et al, serving as a measurement of our own model's effectiveness.

III. Literature Review:

The first paper we evaluated for this project was "Application of Convolutional Neural Networks For Wild Fire Detection" by Khan et al. Khan et al. propose using Convolutional Neural Networks to improve the accuracy of a previously developed low-power, low-cost wildfire detection system. The researchers trained various CNN architectures using a large dataset of wildfire and clean images to classify the presence of fire or smoke. Their goal was to improve upon the limitations of their previous methodology of using a distributed sensor network that was cost-effective in its use of cameras and humidity and smoke sensors but lacked a centralized foundational model. Notably, this model relied upon ground-level imagery as opposed to satellite-based imagery data. For them, this meant they had a direct way to evaluate the performance of their model, but our model will have to operate a bit differently. In addition to differing model architectures, data (and data augmentations), we will use BCEWithLogitsLoss instead of binary-cross entropy (their loss function). This is an industry-best practice for binary classification in modern deep learning because BCEWithLogitsLoss optimizes learning and measures how the model's confidence varies from the truth.

The second and final paper we discovered for this project and the one we will be comparing to our finalized model was "Wildfire Detection Using Convolutional Neural Networks and PRISMA Hyperspectral Imagery: A Spatial-Spectral Analysis" by Spiller et al. This research explores the effectiveness of combining PRISMA hyperspectral satellite imagery with various convolutional neural network architectures to detect wildfires across diverse global ecosystems. The authors trained fully connected, one-dimensional, two-dimensional, and three-dimensional CNN models using data from Australia, and Sicily and then tested the model's generalization abilities on a dataset from Oregon. This is

a more direct comparison to the work that we will be doing, directly using satellite imagery to detect wildfires. We will be using the models built from their paper to serve as a benchmark for our own models.

IV. Methods and Data Sources:

We are using the Wildfire Prediction Dataset (Satellite Images) from [Kaggle](#). The creator generated these images by extracting satellite imagery using the Map Box API at latitude and longitude coordinates of wildfire events (>0.01 acres burned) from the Canadian government wildfire dataset. All images are colored. Original wildfire records come from the Government of Canada.

Description with some basic stats:

The dataset contains satellite images (350px x 350 px) in 2 classes: Wildfire (22710 images) and No wildfire (20140 images). File names are listed as coordinates. The data was divided into Train (~70%), Test (~15%), and Validation (~15%) folders. Within each folder, there are two subfolders wildfire and nowildfire, containing images with wildfire, and no wildfire, respectively. Each image is represented by their geospatial coordinate location.

We performed multiple transformations on our images for our CNN model, including:

- Resizing images to 224px x 224px from 350px x 350 px
- Random horizontal and vertical flips with $p = 0.5$
- Random rotation of 15 degrees
- Color jitter with brightness = 0.1 and contrast = 0.1
- Image normalization based on standard ImageNet stats

Methods:

We are aiming to build a CNN that can effectively generalize satellite images and can accurately predict wildfire or no wildfire images within our test set. Additionally, we will use BCEWithLogitsLoss as our loss function, which is designed for binary classification and includes a parameter that we can adjust to define which incorrect decision we prefer (PyTorch Documentation). Utilizing our results, we plan to visualize model predictions across Canada to identify what areas are more at risk, and how these wildfires are coming in based on the geography and ultimately use that to help prevent and protect the country and areas from wildfires. Lastly, we plan to compare and expand our CNN model, architecture, and results based on prior works (Spiller et al.), to serve as a benchmark for our model's performance.

We propose the following CNN architecture:

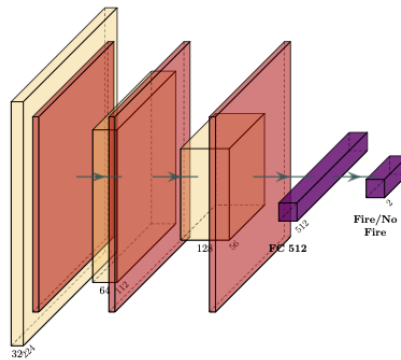


Figure 1: Model Architecture

This includes three convolution blocks, followed by batch normalization, ReLu activation, and max pooling for each individual block, scaled from 32 to 128 channels. After the third convolutional block, we flatten to for our first fully connected layer, then perform dropout with $p = 0.3$ to prevent overfitting, with

our final fully connected layer of fire/no fire to determine whether the image should be classified as wildfire or not.

Note: Our threshold for classifying yes wildfire was lowered from the standard 0.5 to 0.3. In the context of wildfire monitoring, the cost of a False Negative (a missed wildfire) significantly outweighs the operational cost of a False Positive as positives could be flagged for review.

V. Analysis and Interpretation:

With our proposed model architecture, we ran it over 10 epochs yielding the following loss curves over the training and validation data sets. Our model was able to perform well, as the train and validation losses are both low, and we found training over more epochs led to overfitting.

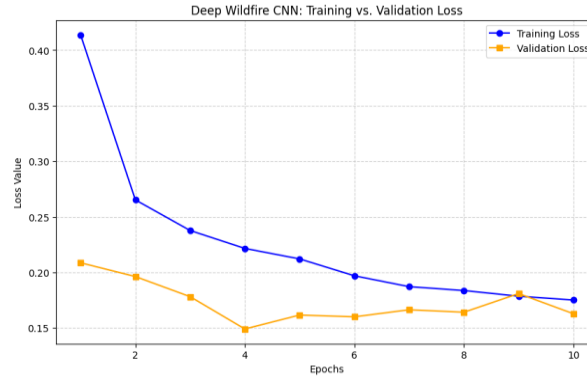


Figure 2: Training vs. Validation Loss for our Final Model

Additionally, utilizing GeoPandas, we were able to plot our results on a map of the Canadian landscape to see the model's confidence and where most of the predicted wildfires ended up. To do this, we converted the filenames from the images (with latitude and longitude) and converted those images to a GeoPandas data frame. The results are plotted below:

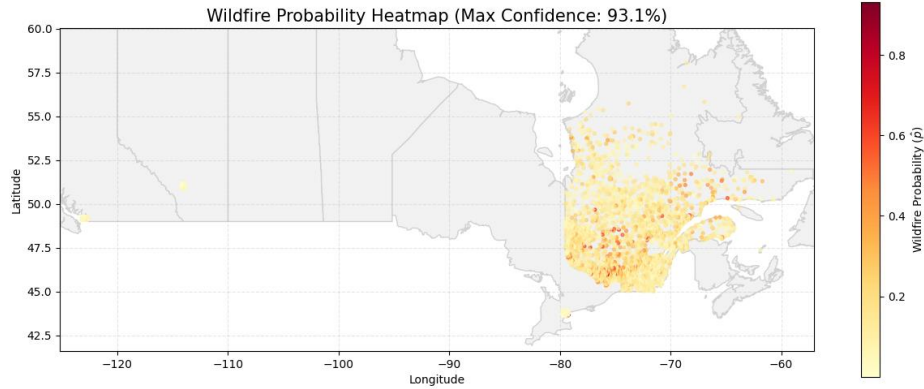


Figure 3: Wildfire probability Map

Comparing our results to the similar models build from Spiller et al, we had to perform an image_to_numpy conversion for their model (as their model expected an array instead of an image). Below compares model performance for our proposed CNN architecture “DEEPWILDFIRECNN” to Spiller et al “IGARSS-STYLECNN”. Based on precision, recall, f1-score, and overall accuracy, our CNN model noticeably outperformed the model outlined by Spiller et al.

--- PYTORCH DEEPWILDFIRECNN ---				
	precision	recall	f1-score	support
No Wildfire	0.97	0.98	0.98	2820
Wildfire	0.98	0.98	0.98	3480
accuracy			0.98	6300
macro avg	0.98	0.98	0.98	6300
weighted avg	0.98	0.98	0.98	6300
--- TENSORFLOW IGARSS-STYLE CNN ---				
	precision	recall	f1-score	support
No Wildfire	0.93	0.94	0.94	2820
Wildfire	0.95	0.95	0.95	3480
accuracy			0.94	6300
macro avg	0.94	0.94	0.94	6300
weighted avg	0.94	0.94	0.94	6300

Figure 4: Table Showing Comparison Between Two Models

Additionally, when looking at f1-scores across decision thresholds, our model performed better than our counterparts at each decision level and significantly outperformed them with decision threshold > 0.7 . This is likely due in part to the model from Spiller et al not having any predictions made above the threshold. However, it does highlight how balanced our CNN model is, across all decision thresholds.

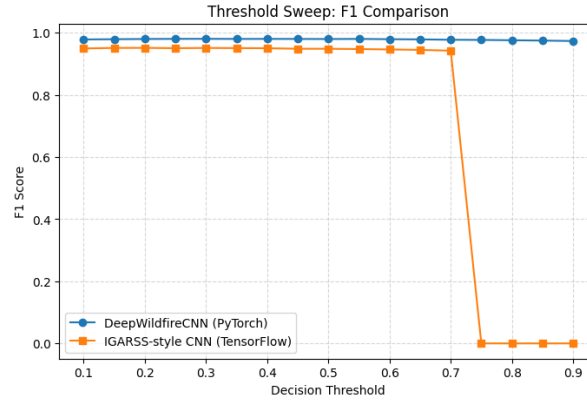


Figure 5: Comparison of two models at different decision thresholds

VI. Discussions/Conclusion

We were able to propose, build, and apply a CNN model that was able to effectively make predictions on unseen data. Not only did our model outperform the model built by Spiller et al, but it possesses real-world applicability, and with a data pipeline with satellite images, our model can be applied and used across the Canadian landscape to help aid predicted wildfires before it becomes too large/harmful. Additionally, we could attempt to apply our model in a similar fashion across similar landscapes – such as the United States, to plan and protect for wildfires across the globe. Additionally, there are a variety of modeling changes that we could make or apply, which include expanding or modifying our own architecture – for example, adding a fourth convolution block, or utilizing dropout more frequently after each block. We could additionally attempt and propose different frameworks, such as object-model detection, prior model architectures for image classification (VGG, ResNet, GoogleNet, etc.). Furthermore, we only trained our model over 10 epochs, it's possible that we could build an even more effective network with small modifications to combat overfitting, and train over many epochs, though it is computationally expensive. Overall, our model is extremely useful and applicable to real-world contexts and scenarios.

References

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